DRUGS · PAIN AND NAUSEA

Antiemetics: ondansetron and others

Post-operative nausea and vomiting is common after ENT surgery, particularly after tonsillectomy (swallowed blood and pain), middle ear surgery (vestibular stimulation), strabismus-like ear operations in children, and in young women and non-smokers. It delays discharge, causes dehydration and, after tonsillectomy, can provoke bleeding. Prevention uses risk scoring and a combination of drugs acting on different receptors.

1 CHAPTER ONE At a glance

The drugs on one page.

1.1 Summary

ANTIEMETICS IN ENT

	SUMMARY
Ondansetron	5-HT3 antagonist; adult 4 mg intravenously at the end of surgery; child 100 micrograms/kg (maximum 4 mg); QT prolongation, headache, constipation
Dexamethasone	Adult 4 to 8 mg at induction; child 150 micrograms/kg (maximum 8 to 10 mg) at tonsillectomy
Cyclizine	H1 antihistamine; 50 mg up to three times daily; useful after ear surgery; tachycardia and dry mouth; sedation
Prochlorperazine	D2 antagonist; 12.5 mg intramuscularly or 3 to 6 mg buccally; extrapyramidal effects
Metoclopramide	D2 antagonist and prokinetic; 10 mg up to three times daily for up to 5 days; extrapyramidal reactions (young women and children; avoid under 1 year, restricted in under-18s); no use in bowel obstruction
Droperidol	D2 antagonist; low dose (0.625 to 1.25 mg) for prophylaxis; QT prolongation

	SUMMARY
Others	Aprepitant (NK1 antagonist) for high-risk patients; hyoscine patch for motion sickness and prolonged vestibular nausea; total intravenous anaesthesia reduces risk

2 CHAPTER TWO Using them in ENT

Practice.

2.1 Assessing risk

The **Apfel score** in adults counts female sex, non-smoking, history of PONV or motion sickness, and post-operative opioid use: each adds about 20% risk. In children, the risk is high after tonsillectomy and strabismus surgery, in those over 3 years, and with a history of PONV.

2.2 Prevention

Give one or two prophylactic agents for moderate risk and three or more for high risk, with different mechanisms; reduce baseline risk with propofol-based anaesthesia, avoidance of nitrous oxide, opioid-sparing analgesia and good hydration. Treat breakthrough vomiting with a drug from a class not already given.

3 CHAPTER THREE Safety

What goes wrong.

3.1 Key cautions

QT prolongation with ondansetron and droperidol (check other QT drugs and electrolytes); dystonic reactions with metoclopramide and prochlorperazine (treat with procyclidine); sedation with cyclizine and hyoscine.

4 CHAPTER FOUR For the examination

What the examiner wants to hear.

4.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- The **Apfel score** and multimodal prophylaxis.
- Why PONV matters after tonsillectomy and middle ear surgery.

4.2 Common pitfalls

COMMON PITFALLS

- Repeating the same antiemetic class for breakthrough vomiting.

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